

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                    |
|----------------------|--------------------|
| United States Patent | 12404919           |
| Kind Code            | B2                 |
| Date of Patent       | September 02, 2025 |
| Inventor(s)          | Wegener; Andreas   |

---

### Linear drive

---

#### Abstract

A linear drive includes an output member which is linearly movable with respect to a drive housing, and which has a spindle nut and a hollow output rod connected thereto. The output member is linearly displaceable by means of a rotationally drivable threaded spindle, which is inserted through the spindle nut into the output rod. To support the threaded spindle, a support apparatus is provided at its anterior end section, which has a rolling bearing and a sliding sleeve seated thereon. The sliding sleeve is divided into sliding sleeve segments, which are connected to one another by resilient joints and are each prestressed with a support area section against the inner peripheral surface of the output rod.

---

|                              |                                              |
|------------------------------|----------------------------------------------|
| <b>Inventors:</b>            | <b>Wegener; Andreas (Stuttgart, DE)</b>      |
| <b>Applicant:</b>            | <b>FESTO SE &amp; CO. KG (Esslingen, DE)</b> |
| <b>Family ID:</b>            | <b>85724543</b>                              |
| <b>Assignee:</b>             | <b>Festo SE &amp; Co. KG (Esslingen, DE)</b> |
| <b>Appl. No.:</b>            | <b>18/847044</b>                             |
| <b>Filed (or PCT Filed):</b> | <b>March 15, 2023</b>                        |
| <b>PCT No.:</b>              | <b>PCT/EP2023/056566</b>                     |
| <b>PCT Pub. No.:</b>         | <b>WO2023/174988</b>                         |
| <b>PCT Pub. Date:</b>        | <b>September 21, 2023</b>                    |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20250207658 A1          | Jun. 26, 2025           |

## Foreign Application Priority Data

DE

10 2022 106 373.9

Mar. 18, 2022

---

## Publication Classification

**Int. Cl.:** F16H25/24 (20060101); F16C27/06 (20060101); F16H25/20 (20060101)

**U.S. Cl.:**

**CPC** F16H25/24 (20130101); F16C27/066 (20130101); F16H2025/204 (20130101); F16H2025/2436 (20130101)

## Field of Classification Search

**CPC:** F16H (25/24); F16H (2025/204); F16H (2025/2436); F16C (19/527); F16C (27/06); F16C (27/066); F16C (31/04); F16C (33/20); F16C (35/12)

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC             |
|--------------|-------------|---------------|----------|-----------------|
| 8495924      | 12/2012     | Fukano        | 74/89.44 | F16H<br>25/2454 |
| 9279450      | 12/2015     | Zaike         | N/A      | F16H 19/04      |
| 9482327      | 12/2015     | Kondo         | N/A      | F16H<br>25/2204 |
| 10919386     | 12/2020     | Seipel        | N/A      | F16F 1/3842     |
| 2013/0248280 | 12/2012     | Stamm et al.  | N/A      | N/A             |
| 2014/0300228 | 12/2013     | Rapp et al.   | N/A      | N/A             |
| 2018/0328472 | 12/2017     | Kuster et al. | N/A      | N/A             |

### FOREIGN PATENT DOCUMENTS

| Patent No.      | Application Date | Country | CPC         |
|-----------------|------------------|---------|-------------|
| 10 2009 007 958 | 12/2011          | DE      | N/A         |
| 10 2013 005 731 | 12/2013          | DE      | N/A         |
| 10 2013 215 865 | 12/2014          | DE      | N/A         |
| 10 2014 213 505 | 12/2015          | DE      | N/A         |
| 102016106237    | 12/2016          | DE      | B62D 5/0403 |
| 2 394 354       | 12/2010          | EP      | N/A         |
| 2 651 745       | 12/2015          | EP      | N/A         |
| 3 401 571       | 12/2017          | EP      | N/A         |
| 6154648         | 12/2016          | JP      | F16H 25/20  |

## OTHER PUBLICATIONS

Examination Report issued in corresponding German Patent Application No. 10 2022 106 373.9, Feb. 8, 2023, 8 pages. cited by applicant

International Search Report issued in International Application No. PCT/EP2023/056566, Aug. 18, 2023, 5 pages w/ translation. cited by applicant  
Written Opinion issued in International Application No. PCT/EP2023/056566, Aug. 18, 2023, 6 pages. cited by applicant

---

*Primary Examiner:* Rushing, Jr.; Bobby

*Attorney, Agent or Firm:* HSML P.C.

---

## **Background/Summary**

(1) This application claims priority to DE 10 2022 106 373 A1, filed 18 Mar. 2022, which is incorporated herein by reference.

### **BACKGROUND OF THE INVENTION**

(2) The invention relates to a linear drive, with a drive housing extending along a housing longitudinal axis, which drive housing encloses a housing interior in which a linearly movable output member extends axially, which output member projects out of the drive housing with an anterior end section usable for force applying and which output member has a hollow output rod in which a rod cavity open at the rear is formed, wherein the output member has a spindle nut which is arranged in the housing interior and is secured against rotation with respect to the drive housing, wherein a threaded spindle passes through the spindle nut, wherein the threaded spindle is in threaded engagement with the spindle nut and has a spindle longitudinal axis extending in the axial direction of the housing longitudinal axis, which threaded spindle, starting from a rear end section of the drive housing, projects through the spindle nut into the rod cavity and can be driven by a rotary drive apparatus, which performs a rotary drive movement about the spindle longitudinal axis, resulting in a linear output movement of the output member relative to the drive housing, wherein an anterior end section of the threaded spindle positioned in the rod cavity is radially supported on an inner peripheral surface of the output rod radially bounding the rod cavity by a support apparatus which has a sliding sleeve coaxial with the threaded spindle, on which sliding sleeve a support area extending in a spindle peripheral direction around the spindle longitudinal axis is formed radially on the outside, along which support area the output rod slides with its inner peripheral surface during its output movement.

(3) A linear drive of this type known from DE 10 2013 005 731 A1 has a drive housing and an output member extending in the drive housing and projecting out of the drive housing at a front side of the housing. The output member has a hollow output rod to which a sleeve-shaped spindle nut is attached at the rear, in which a threaded spindle engages, which can be driven in rotation by an electric motor positioned at the rear of the drive housing. As the spindle nut is secured against rotation with respect to the drive housing, a rotation of the threaded spindle caused by the electric motor results in a linear output movement of the output member. The threaded spindle extends into the hollow output rod and is radially supported in the area of its anterior end section via a support apparatus on the inner peripheral surface of the hollow output rod. The support apparatus consists of a sliding sleeve known as a sliding bush, which has a support area on the radial outside with which it bears against the inner peripheral surface of the hollow output rod with radial support in such a way that the output rod can perform the linear output movement. Tolerance-related radial play in the sliding bearing area between the sliding sleeve and the output rod can cause unwanted running noises and increased wear, as even the slightest imbalance in the drive train can cause the threaded spindle or even the entire linear drive to vibrate. In order to achieve the lowest possible radial play in the area of the sliding sleeve, the sliding sleeves used to the present are usually

selectively provided with suitable dimensions, which involves increased assembly effort and not inconsiderable storage costs. Nevertheless, for functional reasons, a certain radial play cannot be undercut, as the temperatures occurring during operation lead to thermal expansion, which can result in increased friction or even jamming of the components moving relative to each other if the radial play is too low.

(4) A power steering system is known from EP 2 651 745 B1, which has a threaded spindle on which a ball nut is mounted that can rotate relative to a frame, the rotary movement of which results in an axial movement of the threaded spindle. The ball nut is rotatably mounted on the frame by means of a ball bearing. Damping elements are inserted between the outer ring of the ball bearing and the frame.

#### SUMMARY OF THE INVENTION

(5) The problem underlying the invention is to create a linear drive provided with a threaded spindle, which has good running behavior with a simple and cost-effective configuration and causes only low running noise.

(6) In order to solve this problem, for a linear drive of the type mentioned at the beginning, it is provided that the support apparatus has a rolling bearing positioned on the threaded spindle and supporting the sliding sleeve, which rolling bearing allows the threaded spindle to rotate relative to the sliding sleeve, and that the sliding sleeve is divided into a plurality of sliding sleeve segments which are positioned in succession in the spindle peripheral direction and are connected to one another via resilient joints, so that they can be pivoted independently of one another in a radial direction relative to one another and relative to the rolling bearing by a respective swivel area with respect to the spindle longitudinal axis, wherein the support area is formed by support area sections formed on the individual sliding sleeve segments and elastically flexibly prestressed against the inner peripheral surface of the output rod by the resilient joints.

(7) In this way, the threaded spindle is supported radially without play in relation to the output rod, regardless of manufacturing tolerances and temperature fluctuations, in a way that enables smooth linear output movement of the output member equipped with the output rod. The sliding sleeve is segmented in the spindle peripheral direction and divided into individual sliding sleeve segments, each of which has a common support area section forming the support area. Adjacent sliding sleeve segments are connected to each other via resilient joints in such a way that they are elastically prestressed with the support area sections against the inner peripheral surface of the output rod, whereby the resilience enables independent pivoting movements of the sliding sleeve segments in a radial direction with respect to the spindle longitudinal axis in order to maintain constant contact with the inner peripheral surface of the output rod. In addition, the rolling bearing of the support apparatus integrated between the sliding sleeve and the threaded spindle enables the rotational movement of the threaded spindle to be separated from the translational output movement of the output member, so that the relative speed at the sliding points between the output rod and the sliding sleeve segments is reduced, which contributes to a reduction in friction-related wear and operational heat generation.

(8) The sliding sleeve is preferably configured in such a way that the swivel areas of all sliding sleeve segments defined by the resilient joints lie in a common plane orthogonal with respect to the spindle longitudinal axis, which is referred to as the hinge plane for better differentiation. In relation to the longitudinal direction of the threaded spindle, the swivel areas are accordingly at the same axial height.

(9) The resilient joints which are each positioned between two sliding sleeve segments which are adjacent to one another in the spindle peripheral direction are realized in particular as solid state hinges which are integrated in one piece in the sliding sleeve, whereby they each consist of a resiliently twistable bar section of the sliding sleeve which is formed in one piece with the two sliding sleeve segments which are adjacent to one another in the spindle peripheral direction. The function is comparable to that of a torsion spring bar. The material bar forming the bar section,

which is preferably made of the same material as the sliding sleeve segments, is twisted, i.e. torsioned, when the two associated sliding sleeve segments swivel, wherein a restoring force builds up which ensures that the sliding sleeve segments with the support area sections are constantly prestressed against the inner peripheral surface of the output rod, causing the threaded spindle to be radially supported all around at its front end area without play.

(10) A solid state hinge is also referred to as flexure hinge, solid state joint, flexure joint, flexible joint or bending joint.

(11) To form the sliding sleeve segments, the sliding sleeve is preferably provided with longitudinal slots at positions spaced apart from one another in the spindle peripheral direction, so that the sliding sleeve segments adjacent to one another are positioned at a distance from one another except for the resilient joints.

(12) Especially when the resilient joints are realized as solid state hinges, the sliding sleeve can be manufactured particularly cost-effectively from plastic, especially as an injection-molded component.

(13) If the resilient joints are realized as independent components by which sliding sleeve segments configured as separate components are connected to each other, the sliding sleeve segments can also preferably be made of plastic. The use of a thermoplastic material is recommended for any plastic manufacturing.

(14) In principle, the sliding sleeve segments can be configured such that they only project axially from the resilient joints on one side, i.e., in the longitudinal direction of the spindle. Preferred, however, is an embodiment wherein the sliding sleeve segments are configured in a rocker-like manner and can each be swiveled in a rocker-like manner about the associated swivel area in a direction radial to the spindle longitudinal axis due to the resilient joints, wherein each sliding sleeve segment comprises two segment wings extending from its swivel area in opposite axial directions, at least one of which segment wings forms a support wing provided with one of the support area sections. When a rocker-like sliding sleeve segment swivels, one segment wing moves away from the spindle longitudinal axis, while the other segment wing moves closer to the spindle longitudinal axis.

(15) In principle, both segment wings of a respective sliding sleeve segment could comprise a support area section. However, a configuration is considered to be particularly favorable wherein, of the two segment wings of each sliding sleeve segment, only one segment wing forms a support wing provided with a support area section, while the other segment wing is permanently spaced apart from the inner peripheral surface of the hollow output rod and mainly serves to fix the sliding sleeve to the rolling bearing carrying the sliding sleeve, which is why it is referred to as a fastening wing for better differentiation.

(16) In the case of sliding sleeve segments configured as described above, it is preferable if the axial orientation of the two segment wings of the sliding sleeve segments adjacent to each other in the spindle peripheral direction is interchanged in such a way that the support wings point in axially opposite directions starting from the swivel area of the relevant sliding sleeve segment. The same applies to the fastening wings. In this case, the sliding sleeve is presented with two segment wing circles positioned axially next to each other in the longitudinal direction of the spindle, each consisting of support wings and fastening wings positioned alternately in succession in the spindle peripheral direction. The support area sections thus extend in particular in a meandering course around the spindle longitudinal axis.

(17) On its radial inner circumference, the sliding sleeve is preferably provided with an annular seating area which is coaxial with the spindle longitudinal axis, points radially inwards towards the spindle longitudinal axis and with which the sliding sleeve is seated on a radial outer peripheral surface of an outer ring of the rolling bearing fixed to the threaded spindle by an inner ring. The radial outer peripheral surface of the outer ring of the rolling bearing is preferably circular-cylindrical in shape, with the annular seating area in particular having a complementary shape in

this respect. In this way, the sliding sleeve is very effectively supported radially by the rolling bearing.

(18) In particular, the sliding sleeve is configured such that its resilient joints are at the same axial height as the annular seating area, wherein the sliding sleeve segments project axially beyond the annular seating area, in particular axially on both sides in each case.

(19) Preferably, the annular seating area of the sliding sleeve comprises a plurality of segment seating area sections positioned in succession in the spindle peripheral direction, each of which is formed on the radial inner surface of one of the sliding sleeve segments. These segment seating area sections are each bounded axially on only one side by a retaining surface, which is formed on a retaining projection formed on the associated fastening wing and projecting radially inwards. The retaining surface formed on each sliding sleeve segment, which is oriented in the longitudinal direction of the spindle, lies axially against one of the two axial end faces of the outer ring of the rolling bearing. In this way, the outer ring is held between retaining surfaces positioned at a distance from each other in the spindle peripheral direction and facing axially towards the segment seating area sections. The sliding sleeve thus has an annular fastening recess on its inner circumference, the base surface of which is formed by the annular seating area, and which is bounded laterally by a respective recess side surface, which is formed by a plurality of retaining surfaces positioned at a distance from one another in the spindle peripheral direction, the retaining surfaces of the two recess side surfaces being set with a gap between them.

(20) The annular seating area of the sliding sleeve is preferably composed jointly of, on the one hand, the segment seating area sections positioned in succession at a distance in the spindle peripheral direction and, on the other hand, a number of hinge seating area sections corresponding to the number of resilient joints, the hinge seating area sections each being positioned between two successive segment seating area sections and each being formed on the radial inner surface of one of the resilient joints.

(21) The sliding sleeve segments are preferably positioned in an inclined orientation with respect to the spindle longitudinal axis. In relation to a segment longitudinal axis of each sliding sleeve segment, which extends in a plane overlapping with the spindle longitudinal axis, in which the sliding sleeve segment is swivelable, and which can therefore be referred to as the swivel plane, the sliding sleeve segments are accordingly positioned such that the segment longitudinal axes are inclined with respect to the spindle longitudinal axis. Sliding sleeve segments adjacent to each other in the spindle peripheral direction or their segment longitudinal axes are preferably inclined in opposite directions.

(22) If one considers the sliding sleeve in an initial state before insertion into the rod cavity, not mounted in its position of use, the segment longitudinal axes of the sliding sleeve segments are in each case more inclined than in the mounted position of use, in which they are pivoted into a state of lesser inclination by contact with the inner peripheral surface of the output rod, overcoming a spring-loaded restoring force. The spring force with which the sliding sleeve segments are elastically prestressed against the inner peripheral surface of the output rod results from the restoring force that builds up.

(23) Each sliding sleeve segment has a radial segment outer surface facing the inner peripheral surface of the output rod, which in each case forms one of the support area sections. Preferably, the segment outer surface is convexly curved in the area of the support area section formed by it and preferably in its entirety both in the spindle peripheral direction and in the axial direction of the spindle longitudinal axis. The radius of curvature in the spindle peripheral direction corresponds at least essentially to the radius of the inner peripheral surface of the output rod. The curvature in the axial direction of the spindle longitudinal axis ensures that the outer segment surface is in contact with the cylindrical inner peripheral surface of the output rod, regardless of the current swivel position of the sliding sleeve segment.

(24) Preferably, the sliding sleeve segments are each positioned tiltable on the rolling bearing to

enable their radial swiveling. This possibility of tilting is favored in a fastening recess provided with axial retaining surfaces by the fact that the retaining surfaces are only present axially on one side of each sliding sleeve segment. The swiveling and the tilting are preferably one and the same movement.

(25) The axial arrangement of retaining surfaces on only one side to create a mounting recess suitable for holding the outer ring of a rolling bearing also has the advantage that the sliding sleeve can be axially demolded very easily during injection molding production.

(26) Preferably, the linear drive comprises a rotary drive apparatus positioned at the rear of the drive housing, which is coupled to the threaded spindle in a torque-transmitting manner to generate its drive rotary movement.

(27) The rotary drive apparatus can certainly be configured as a fluid-actuated rotary drive apparatus. Preferably, however, an electric rotary drive apparatus is used and, in particular, an electric motor, with which very high speeds and consequently high traversing speeds of the output member can be realized. Preferably, the electric motor is a stepper motor that enables simple and reliable axial positioning of the output member in relation to the drive housing if required.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention is explained in more detail below with reference to the enclosed drawing. It shows:

(2) FIG. 1 a preferred embodiment of the linear drive according to the invention in a longitudinal section according to sectional plane I-I from FIG. 2, wherein the area of the support apparatus is illustrated separately, also enlarged again, and furthermore the sliding sleeve is shown in a further image section alone,

(3) FIG. 2 a cross-section of the linear drive according to sectional plane II-II from FIG. 1, and

(4) FIG. 3 a perspective view of the sliding sleeve included in the linear drive of FIGS. 1 and 2.

### DETAILED DESCRIPTION

(5) The linear drive 1 has a housing referred to as drive housing 2 with a central housing longitudinal axis 3 indicated by a dotted line, the axial direction of which is referred to as housing longitudinal direction 3a.

(6) The drive housing 2 extends in the longitudinal direction of the housing 3a between a front side of the housing 4 and a rear side of the housing 5. The drive housing 2 encloses a housing interior 7.

(7) Exemplarily, the drive housing 2 contains a front housing cover 8 positioned at the front of the housing 4, a rear housing cover 9 positioned at the rear of the housing 5 and a housing tube 6 extending between them in the longitudinal direction of the housing 3a.

(8) Preferably, the front and rear housing covers 8, 9 are attached to the respective end face of the housing tube 7 and axially braced to the housing tube 6 by means of several fastening screws. As an example, the housing tube 6 has a plurality of fastening holes 13, as shown in FIG. 2, into which the fastening screws engage.

(9) The housing tube 6 has a radially inner tube circumferential surface 14 peripherally delimiting the housing interior 7, which preferably extends in the longitudinal direction of the housing 3a over the entire length of the tube.

(10) The linear drive 1 also has a multi-part output member 15 with a longitudinal shape. The output member 15 extends in the housing interior 7 in the longitudinal direction 3a, preferably being positioned coaxially to the housing tube 6.

(11) The output member 15 has a rear end section 16 positioned in the housing interior 7 and facing the rear of the housing 5 and a front end section 17 projecting out of the drive housing 2 at the front of the housing 4. The output member 15 passes through the front housing cover 8 as an example,

being enclosed by a guide bush **10** fixed in the front housing cover **8** and guided for linear displacement with radial support.

(12) When the linear drive **1** is used, the output member **15** can be driven in two opposite axial directions relative to the drive housing **2** in a linear output movement **18** indicated by a double arrow. The anterior end section **17** of the output member **15**, which is always located outside the drive housing **2**, can be used to apply force, whereby it preferably has a force-applying structure **19** to which any component to be moved can be fixed, for example a machine element to be moved.

(13) Depending on the direction of movement, the linear output movement **18** manifests itself in an extension or retraction of the output member **15** with respect to the drive housing **2**.

(14) The output member **15** includes a spindle nut **22** positioned in the housing interior **7**, which is secured against rotation with respect to the drive housing **2**. The anti-rotation lock prevents the spindle nut **22** from rotating relative to the drive housing **2** about the housing longitudinal axis **3**. However, the spindle nut **22** can easily follow the linear output movement **18**.

(15) The anti-rotation lock of the spindle nut **22** is exemplarily realized in that a radially protruding anti-rotation projection **23** is positioned radially on the outside of the spindle nut **22**, which is inserted into an anti-rotation groove **24** of the drive housing **2** provided with a complementary cross-section. The anti-rotation groove **24** is formed on the inner circumferential surface **14** of the tube and extends as a longitudinal groove between the two housing covers **8, 9**, wherein it has a slot-like groove opening **25** facing the housing interior **7**. The anti-rotation projection **23** is inserted through the groove opening **25** into the anti-rotation groove **24** and can move in the longitudinal direction of the anti-rotation groove **24** during the output movement **18**. The support of the anti-rotation projection **23** on the two groove flanks of the anti-rotation groove **24** results in the aforementioned anti-rotation of the spindle nut **22**. For example, the anti-rotation projection **23** is ridge-shaped.

(16) According to an embodiment not illustrated, the anti-rotation lock of the spindle nut **22** can also be realized by the housing interior **7** having a non-circular cross-section and the spindle nut **22** having a complementary cross-sectional contour in this respect.

(17) The spindle nut **22** is axially penetrated centrally by a through opening **26**, which is provided with an internal thread **27** on its inner circumference. The internal thread **27** is preferably realized with a recirculating ball principle and includes a plurality of rolling bearing balls that can circulate in a closed ball channel system, which consists of a helical groove in the area of the inner circumference of the through opening **26**, from which the rolling bearing balls partially protrude. In principle, however, the internal thread **27** can also be realized in another way.

(18) The output member **15** further comprises a hollow output rod **28**, which extends coaxially to the housing longitudinal axis **3** and passes through the front housing cover **8** in an axially displaceable manner in this respect. An outer end section **30** of the hollow output rod **28** located outside the drive housing **2** forms the front end section **17** of the output member **15** and, by way of example, carries the force-applying structure **19**. An inner end section **31** of the output rod **28** is located in the housing interior **7** and is attached there to the spindle nut **22** in such a way that a cavity extending axially in the output rod **28**, referred to as rod cavity **32**, is aligned with the through opening **26** of the spindle nut **22**.

(19) Since the rod cavity **32** is open at the rear, i.e., at the inner end section **31**, the through opening **26** and the rod cavity **32** merge axially into one another.

(20) The spindle nut **22** is coupled to the output rod **28** in any desired way in the longitudinal direction of the housing **3a**. This means that the output rod **28** follows every movement of the spindle nut **22** in the longitudinal direction of the housing **3a**. For example, the two parts are welded or pressed together.

(21) The linear drive **1** has a rotary drive apparatus **33**, which is preferably positioned in the area of the rear of the housing **5** on the drive housing **2** and, in particular, is attached to it. The rotary drive apparatus **33** is only indicated schematically in the drawing.



(22) The rotary drive apparatus **33** is connected in terms of drive to a threaded spindle **34**, which projects axially into the housing interior **7** starting from a rear end section of the drive housing **2**, which is formed by the rear housing cover **9** as an example. The threaded spindle **34** has a spindle longitudinal axis **37**, the axial direction of which is referred to as spindle longitudinal axis **37a** and which preferably has the same orientation as the housing longitudinal axis **3**. Preferably, the spindle longitudinal axis **37** and the housing longitudinal axis **3** overlap, which applies to the embodiment shown.

(23) The threaded spindle **34** has a rear end section **35**, which is rotatably mounted via a rotary bearing apparatus **36** of any type in the rear end section of the drive housing **2**, formed for example by the rear housing cover **9**, so that it can rotate relative to the drive housing **2** about its spindle longitudinal axis **37** and at the same time is radially supported. The pivot bearing apparatus **36** is preferably configured as a roller bearing apparatus.

(24) The rotary drive apparatus **33** is connected to the threaded spindle **34** in a torque-transmitting manner, which is exemplarily realized in that the threaded spindle **34** has a torque introduction section **38** on the rear side, which is connected to a drive shaft of the rotary drive apparatus **33** via a coupling apparatus.

(25) The rotary drive apparatus **33** enables the threaded spindle **34** to be driven rotationally in one of two opposing directions of rotation to a drive rotary movement **42** about the spindle longitudinal axis **37**, indicated by a double arrow.

(26) Preferably, the rotary drive apparatus **33** is of an electrical type, wherein it is preferably configured as an electric motor, which applies to the illustrated embodiment example. In particular, the rotary drive apparatus **33** is an electric stepper motor.

(27) The threaded spindle **34** extends through the through opening **26** of the spindle nut **22** and protrudes into the rod cavity **32** of the output rod **28**. The threaded spindle **34** has an external thread **43** on its outer circumference, which is in threaded engagement with the internal thread **27** of the spindle nut **22**.

(28) Preferably, the threaded spindle **34** extends in the housing interior **7** at least approximately into the area of the exemplary front end section of the drive housing **2** formed by the front housing cover **8**.

(29) One direction around the spindle longitudinal axis **37** is referred to as the spindle peripheral direction **44** and is indicated in the drawing by a double arrow. The drive rotary movement **42** takes place in the spindle peripheral direction **44** either clockwise or counterclockwise. Due to the threaded engagement of the output rod **28** with the spindle nut **22**, which is secured against rotation with respect to the drive housing **2**, a drive rotary movement **42** of the threaded spindle **34** results in the linear output movement **18** of the output member **15** already mentioned above. The direction of movement of the output movement **18** depends on the direction of rotation of the drive rotary movement **42**.

(30) The output rod **28** has an anterior end section **45** which ends freely in the rod cavity **32** and is supported by a support apparatus **46** of the linear drive **1** on the inner peripheral surface **47** of the hollow output rod **28**, which radially delimits the rod cavity **32**, in a radial direction with respect to the spindle longitudinal axis **37**, without impeding the linear displaceability of the output member **15** and the rotational movement of the threaded spindle **34** required with respect to the output member **15**.

(31) The support apparatus **46** includes a sliding sleeve **48** and an annular rolling bearing **49**. The rolling bearing **49** is seated coaxially on the anterior end section **45** of the threaded spindle **34** and carries the sliding sleeve **48**, which is also positioned coaxially to the threaded spindle **34**. A support area **52** is formed radially on the outside of the sliding sleeve **48**, which extends in the spindle peripheral direction **44** around the spindle longitudinal axis **37** and bears against the inner peripheral surface **47** of the output rod **28**. During the output movement **18**, the output rod **28** slides with its inner peripheral surface **47** against the support area **52**.

- (32) Due to an advantageous structure of the sliding sleeve **48**, which will be explained below, its support area **52** is constantly resiliently prestressed against the inner peripheral surface **47**. This results in radially clearance-free support of the anterior end section **45** of the threaded spindle **34** on the output rod **28**. As a result, the rotation of the threaded spindle **34** is very smooth while undesirable vibrations are avoided.
- (33) The rolling bearing **49** enables the threaded spindle **34** to rotate relative to the sliding sleeve **48**. Relative movements between the sliding sleeve **48** and the output rod **28** are therefore predominantly limited to translational movements in the longitudinal direction of the housing **3a** with no or only slightly superimposed rotational movements in the spindle peripheral direction **44**. This separation of rotational and translational movements reduces the relative speed at the contact areas between the support area **52** and the inner peripheral surface **47**, which also contributes to a reduction in running noise and also minimizes friction-related wear.
- (34) The rolling bearing **49** has an inner ring **53**, an outer ring **54** coaxially surrounding the inner ring **53** with radial spacing and a plurality of rolling elements **55** positioned in a rolling manner between the inner ring **53** and the outer ring **54**. In particular, this is a radial rolling bearing in which the rolling elements **55** enable radial force transmission between the inner ring **53** and the outer ring **54**.
- (35) As an example, the rolling bearing **49** is configured as a ball bearing with rolling balls as rolling elements **55**. The rolling bearing **49** can also be configured as a roller bearing with rolling rollers as rolling elements **55**.
- (36) The rolling bearing **49** is seated with its inner ring **53** on the anterior end section **45** of the output rod **28**. The output rod **28** ends, by way of example, with a cylindrical bearing extension **56**, which is coaxially enclosed by the inner ring **53**. Preferably, the rolling bearing **49** is fixed with its inner ring **53** in a press fit on the bearing extension **56**. Other fastening measures, for example axially immovable fastening by means of a retaining ring, are also possible.
- (37) The outer ring **54** has a preferably cylindrical radial outer peripheral surface **57** and axially on both sides of this outer peripheral surface **57** one of two oppositely oriented annular axial end faces **58a**, **58b**. The outer ring **54** can rotate around the inner ring **53**, with the rolling elements **55** rolling on the inner peripheral surfaces of the inner ring **53** and the outer ring **54**.
- (38) An annular groove is formed in the outer circumference of the inner ring **53** and in the inner circumference of the outer ring **54**, into which the rolling elements **55** are partially inserted, wherein the resulting axial form fit has the effect that the outer ring **54** is fixed axially immovable with respect to the inner ring **53**.
- (39) The sliding sleeve **48** has a central longitudinal sleeve axis **61**, which overlaps with the spindle longitudinal axis **37** in the operational position of use of the sliding sleeve **48**. A sleeve peripheral direction **62**, illustrated by the same double arrow as the spindle peripheral direction **44**, runs around the sleeve longitudinal axis **61** and thus also around the spindle longitudinal axis **37**.
- (40) The sliding sleeve **48** has an inner peripheral surface **63** facing the sleeve longitudinal axis **61** and an outer peripheral surface **64** facing away radially with respect to the sleeve longitudinal axis **61**. An annular fastening recess **65** coaxial with the sleeve longitudinal axis **61** is formed on the inner peripheral surface **63**, by means of which the sliding sleeve **48** is seated on the outer ring **54**. At least a section of the radial height of the outer ring **54** is inserted into the fastening recess **65**, so that the outer ring **54** and the sliding sleeve **48** are fixed immovably relative to one another in the longitudinal direction of the spindle **37a**.
- (41) The fastening recess **65** has a radially inwardly pointing circular base surface, referred to in the following as seating area **66**, which extends uninterruptedly around the longitudinal axis **61** of the sleeve. The seating area **66** encloses the radial outer peripheral surface **57** of the outer ring **54** and is supported radially thereon.
- (42) Laterally, the fastening recess **65** is limited by two recess side surfaces **67a**, **67b** facing each other, each of which is supported on one of the two axial end faces **58a**, **58b** of the outer ring **54**. In

this way, the sliding sleeve **48** is fixed axially immovably to the outer ring **54** of the rolling bearing **49**.

(43) The two recess side surfaces **67a**, **67b** are each segmented in the sleeve circumferential direction **62** and are formed of a plurality of circular arc-shaped side surface portions which follow each other at a distance in the sleeve circumferential direction **62** and are hereinafter referred to as retaining surfaces **68**. Opposite each retaining surface **68** of each recess side surface **67a**, **67b** is an area of the respective other recess side surface **67b**, **67a** which has no retaining surface and is thus axially open in the axial direction of the sleeve longitudinal axis **61**.

(44) The outer ring **54** is preferably clamped between the retaining surfaces **68** of the two recess side surfaces **67a**, **67b**. Normally, therefore, no rotational relative movements take place between the sliding sleeve **48** and the outer ring **54** during operation of the linear drive **1**. The threaded spindle **34** can therefore rotate relative to the sliding sleeve **48** held on the outer ring **54** during its drive movement **42**.

(45) Depending on the radial contact pressure prevailing between the support area **52** and the inner peripheral surface **47** of the output rod **28**, on the intensity of the clamping of the sliding sleeve **48** on the outer ring **54** and on the internal friction of the rolling bearing **49**, the rotating threaded spindle **34** can possibly also cause a rotation of the sliding sleeve **48** about the spindle longitudinal axis **37**, so that not only translational relative movements in the longitudinal spindle direction **37a** but also rotational relative movements in the spindle peripheral direction **44** occur between the sliding sleeve **48** and the output rod **28**, although the rotational speed of the sliding sleeve **48** is considerably lower than that of the threaded spindle **34**. An effort will be made to configure the design in such a way that the sliding sleeve **48** performs no or at least only minimal rotational movements relative to the output rod **28**.

(46) Insofar as the sleeve peripheral direction **62** is referred to in the following and no deviating information is provided in individual cases, this also refers to the spindle peripheral direction **44**, and vice versa. The same applies to the spindle longitudinal axis **37** on the one hand and the sleeve longitudinal axis **61** on the other.

(47) A special feature of the structure of the sliding sleeve **48** is that it is segmented in the spindle peripheral direction **44**, which is expressed by the fact that it is subdivided in the spindle peripheral direction **44** or the sleeve peripheral direction **62** into a plurality of sliding sleeve segments **71** positioned in succession. However, the sliding sleeve **48** nevertheless is a coherent ring structure, since adjacent sliding sleeve segments **71** in the sleeve circumferential direction **62** are each connected to one another via a resilient joint **72** of the sliding sleeve **48**. These resilient joints **72** enable the individual sliding sleeve segments **71** to swivel independently of one another both relative to one another and relative to the rolling bearing **49** in a radial direction with respect to the spindle longitudinal axis **37**. The possible swivel movement **73** of the sliding sleeve segments **71**, which is also referred to below as segment swivel movement **73**, is indicated in the drawing by double arrows.

(48) During the segment swivel movement **73**, the sliding sleeve segment **71** in question swivels about a swivel area **74**, which could also be referred to as the swivel center and which is defined by the two resilient joints **72**, on which each sliding sleeve segment **71** is resiliently suspended.

(49) The segment swivel movement **73** takes place for each sliding sleeve segment **71** in a swivel plane **75** indicated by a dotted line, which is spanned by the spindle longitudinal axis **37** and a radial axis passing through the relevant sliding sleeve segment **71**.

(50) The spring elasticity of the joints **72** results, by way of example, from their configuration as elastically deformable solid state hinges **72a**. For this purpose, adjacent sliding sleeve segments **71** are separated from one another in the spindle peripheral direction **44** by means of radially continuous longitudinal slots **76**, whereby these longitudinal slots **76** do not, however, completely penetrate the sliding sleeve **48** axially, so that a narrow bar section **77** remains, which is formed in one piece and connects the two adjacent sliding sleeve segments **71** to one another in one piece.

This bar section **77** defines a material bar which has the properties of a torsion spring bar and can be twisted about a torsion axis which is aligned in the sleeve circumferential direction **62**.

(51) By way of example, the bar sections **72** each have an at least substantially rectangular cross-section.

(52) The sliding sleeve **48** is therefore preferably formed as a single piece, whereby it is composed of sliding sleeve segments **71** and solid state hinges **72a** that merge into one another as a single material. Preferably, the sliding sleeve **48** is made of a plastic and, in particular, of a thermoplastic.

(53) Preferably, all resilient joints **72** lie together in a plane **78** orthogonal to the spindle longitudinal axis **37**, which is referred to as hinge plane **78** for better differentiation. Accordingly, the swivel areas **74** of all sliding sleeve segments **71** also lie in said hinge plane **78**.

(54) According to an illustrated preferred embodiment, the sliding sleeve segments **71** are configured in a rocker-like manner and are suspended by the resilient joints **72** in such a way that they are each pivoted in a rocker-like manner when the segment swiveling movement **73** is executed. Each sliding sleeve segment **71** has a first segment wing **71a** and a second segment wing **71b**, these two segment wings **71a**, **71b** projecting in opposite axial directions of the sleeve longitudinal axis **61** starting from the swivel area **74** in which the resilient joints **72** engage and each ending freely. One of the longitudinal slots **76** extends between each of the resilient joints **72** and the free ends of the segment wings **71a**, **71b**. During the segment swiveling movement **73**, the respective segment wing **71a** or **71b** approaches the spindle longitudinal axis **37**, while at the same time the respective other segment wing **71b**, **71a** moves away from the spindle longitudinal axis **37**. Different swivel positions of the sliding sleeve segments **71** are strongly exaggerated in the enlarged individual representation of the sliding sleeve **48** by means of dashed contours.

(55) The support area **52** is also segmented in the spindle peripheral direction **44** in accordance with the segmentation of the sliding sleeve **48**. The support area **52** is formed from a number of support area sections **79** corresponding to the number of sliding sleeve segments **71**, each of which is formed on one of the sliding sleeve segments **71**. Accordingly, the support area **52** is not a closed annular area but is interrupted in each of the joint areas of the sliding sleeve **48**.

(56) Each individual support area section **79** is elastically flexibly pressed against the inner peripheral surface **47** of the output rod **28** with a spring force due to the spring-elastic effect of the solid state hinges **72a** present as an example.

(57) The resilient compressive force results, by way of example, from a structure of the sliding sleeve **48** such that the sliding sleeve segments **71** are pressed radially inwards against a resilient restoring force of the resilient joints **72** when the sliding sleeve **48** is inserted into the rod cavity **32** from a stress-neutral initial position shown in FIG. 3. In the stress-neutral initial state with the sliding sleeve **48** not mounted, the outer area sections **79** lie on a circle with a larger diameter compared to the rod cavity **32**, so that they are pivoted inwards when the sliding sleeve **48** is pushed into the rod cavity **32**, building up a rebounding support force.

(58) To explain this effect, reference is made to the fact that each sliding sleeve segment **71** has a segment longitudinal axis **82** indicated by a dotted line, which extends, by way of example, between the free end regions of the two segment wings **71a**, **71b** and which lies in the above-mentioned swivel plane **75** of the relevant sliding sleeve segment **71**. These segment longitudinal axes **82** are inclined with respect to the sleeve longitudinal axis **61**, wherein they form an acute angle with the sleeve longitudinal axis **61** in the associated swivel plane **75**. In the unassembled initial position according to FIG. 3, the segment longitudinal axes **82** are more inclined than in the assembled position of use according to FIG. 1. This change in inclination causes the desired resilient support of the sliding sleeve segments **71** on the inner peripheral surface **47** of the hollow output rod **28**.

(59) Preferably, the support area sections **79** do not extend over the entire length of the respectively assigned sliding sleeve segment **71**, but are each limited to one of the two segment wings **71a**, **71b**. The segment wing **71a** or **71b** having the support area section **79** is hereinafter also referred to as

support wing **83**.

(60) The segment wing **71a** or **71b**, which does not function as support wing **83**, is always radially spaced apart from the inner peripheral surface **47** of the hollow output rod **28**. The section of the outer peripheral surface **64** of the sliding sleeve **48** formed on it has no contact with the inner peripheral surface **47**. This results from the fact that the sliding sleeve segments **71** are inclined with respect to the longitudinal axis **61** of the sleeve not only in the initial position, but also in the position of use of the sliding sleeve **48**.

(61) The segment wings **71a** or **71b**, which are spaced apart from the inner peripheral surface **47** and do not perform a supporting function, are referred to below as fastening wings **84**, as they play a decisive role in fastening the sliding sleeve **48** to the outer ring **54** of the rolling bearing **49**.

(62) Starting from the swivel area **74**, all first segment wings **71a** project in the same axial direction, namely, by way of example, axially away from the threaded spindle **34** in the direction of the front side of the housing **4**. All second segment wings **71b** are oriented in the opposite direction and project in the direction of the threaded spindle **34** and thus in the direction of the rear side of the housing **5**. The sliding sleeve **48** thus has two segment wing circles **85a**, **85b** positioned next to one another in the longitudinal direction of the spindle **37a**, a first segment wing circle **85a** being formed by the first segment wings **71a** and a second segment wing circle **85b** being formed by the second segment wings **71b**.

(63) A preferred special feature of the sliding sleeve **48** realized in the embodiment is that the axial orientation of the sliding sleeve segments **71** directly adjacent to each other in the spindle peripheral direction **44** is reversed in such a way that the support wings **83** and the fastening wings **84** point in axially opposite directions. In other words, both the first segment wing circle **85a** and the second segment wing circle **85b** are formed from support wings **83** and fastening wings **84** positioned alternately in the spindle peripheral direction **44**. Among other things, this also has the consequence that the support area sections **79** of respectively adjacent sliding sleeve segments **71** in the spindle peripheral direction **44** are offset relative to one another in the spindle longitudinal direction **37a**. This is accompanied by an opposing inclination of the segment longitudinal axes **82** of the sliding sleeve segments **71** directly following each other in the spindle peripheral direction **44** with respect to the spindle longitudinal axis **37**.

(64) The relevant first or second segment wings **71a**, **71b** obtain their functionality as fastening wings **84** from the fact that the retaining surfaces **68** described above, which contribute to the formation of the fastening recess **65**, are formed on them. The interchanged axial orientation of adjacent sliding sleeve segments **71** described above means that the retaining surfaces **68** are positioned with a gap in the spindle peripheral direction **44** and are oriented in opposite directions to one another. Each retaining surface **68** is located on a radially inwardly projecting retaining projection **86** of the fastening wing **84**.

(65) The annular seating area **66** is composed of segment seating area sections **87** formed radially on the inside of the individual sliding sleeve segments **71**, each of which is axially bounded on one side by one of the retaining surfaces **68**, and of hinge seating area sections **88** formed on the radially inner surface of the resilient joints **72** or the solid state hinges **72a**.

(66) Preferably, the resilient joints **72** are at the same axial height with the annular seating area **66** in relation to the axial direction of the sleeve longitudinal axis **61**, with the sliding sleeve segments **71** projecting axially with their segment wings **71a**, **71b** beyond the annular seating area **66**.

(67) The retaining projections **86** have, on their radially inner side axially adjacent to the retaining surface **68**, preferably an insertion slope **89** extending obliquely radially outwards in the direction of the free end of the fastening wing **84**. Together, the insertion slopes **89** of both segment wing circles **85a**, **85b** each form an insertion region tapering conically from the axial outside in the direction of the fastening recess **65**, which could also be described as an insertion funnel. To mount it on the rolling bearing **49**, the sliding sleeve **48** can be pushed onto the rolling bearing **49** with one of the insertion areas in front, wherein the insertion slopes **89** slide along the outer ring **54**,

whereby the sliding sleeve segments **71** are pressed outwards with a segment swivel movement **73** until the rolling bearing **49** has passed the associated retaining surfaces **68**. The sliding sleeve segments **71** then snap back again and the outer ring **54** is fixed between the retaining surfaces **68**. The sliding sleeve **48** can therefore be very easily snapped onto the rolling bearing **49** in an axial insertion process.

(68) The sliding sleeve segments **71** can automatically align themselves in terms of inclination according to the inner diameter of the rod cavity **32** by performing corresponding segment swivel movements **73** during the drive movement **42** of the threaded spindle **34**. The segment swivel movements **73** each affect the entire sliding sleeve segment **71**, which has a corresponding rigidity. Since this may be accompanied by tilting with respect to the outer ring **54**, the actual contact area between the seating surface **66** and the radial outer peripheral surface **57** of the outer ring **54** generally also changes during the segment swivel movement **73**.

(69) Each sliding sleeve segment **71** has a radial outer segment surface **92** facing the inner peripheral surface **47** of the hollow output rod **28**. Each support area section **79** is formed by a partial area of the associated outer segment surface **92**.

(70) The segment outer surfaces **92** are each convexly curved in the spindle peripheral direction **44** at least in the region of the support area sections **79**, the curvature corresponding to the curvature of the inner peripheral surface **47**, so that the sliding sleeve segments **71** can bear against the inner peripheral surface **47** along their entire width extending in the spindle peripheral direction **44**. Preferably, the curvatures of the inner peripheral surface **47** and the segment outer surfaces **92** are complementary to each other.

(71) Preferably, the segment outer surfaces **92** are convexly curved not only in the spindle peripheral direction **44**, but also in the spindle longitudinal direction **37a**, at least in the area of the support area sections **79**. This preferably results in a convex surface shape. This measure realized in the embodiment has the effect that the support area sections **79** are always in relatively large contact with the inner peripheral surface **47**, regardless of the current swivel position of the sliding sleeve segments **71**. During the segment swivel movement **73**, the current contact area between the inner peripheral surface **47** and the support area sections **79** can move along the segment outer surface **92** in the longitudinal spindle direction **37a**. As already mentioned, the segment outer surfaces **92** can each have a spherical contour overall. It is preferred to intentionally omit a cylindrical contour.

(72) Since the contact points between the sliding sleeve segments **71** and the inner peripheral surface **47** are offset from each other in the longitudinal direction of the spindle **37a**—they are located axially on both sides of the resilient joints **72**—, wherein the sliding sleeve segments **71** are each spring-loaded to the inner peripheral surface **47** of the hollow output rod **28** with no clearance, this results in a very effective radial support of the anterior end section **45** of the threaded spindle **34** regardless of the current axial position of the output member **15**.

## Claims

1. A linear drive, comprising: a drive housing extending along a housing longitudinal axis, which drive housing encloses a housing interior in which a linearly movable output member extends axially, which output member projects out of the drive housing with an anterior end section usable for force applying and which output member has a hollow output rod in which a rod cavity open at the rear is formed, wherein the output member has a spindle nut which is arranged in the housing interior and is secured against rotation with respect to the drive housing, wherein a threaded spindle passes through the spindle nut, wherein the threaded spindle is in threaded engagement with the spindle nut and has a spindle longitudinal axis extending in the axial direction of the housing longitudinal axis, which threaded spindle, starting from a rear end section of the drive housing, projects through the spindle nut into the rod cavity and configured to be driven by a rotary drive

apparatus, which performs a rotary drive movement about the spindle longitudinal axis, resulting in a linear output movement of the output member relative to the drive housing, wherein an anterior end section of the threaded spindle positioned in the rod cavity is radially supported on an inner peripheral surface of the output rod radially bounding the rod cavity by a support apparatus which has a sliding sleeve coaxial with the threaded spindle, on which sliding sleeve a support area extending in a spindle peripheral direction around the spindle longitudinal axis is formed radially on the outside, along which support area the output rod slides with its inner peripheral surface during its output movement, wherein the support apparatus has a rolling bearing positioned on the threaded spindle and supporting the sliding sleeve, which rolling bearing allows the threaded spindle to rotate relative to the sliding sleeve, and wherein the sliding sleeve is divided into a plurality of sliding sleeve segments which are positioned in succession in the spindle peripheral direction and are connected to one another via resilient joints, so that they can be pivoted independently of one another in a radial direction relative to one another and relative to the rolling bearing by a respective swivel area with respect to the spindle longitudinal axis, wherein the support area is formed by support area sections formed on the individual sliding sleeve segments and elastically flexibly prestressed against the inner peripheral surface of the output rod by the resilient joints.

2. The linear drive according to claim 1, wherein the swivel areas of all sliding sleeve segments defined by the resilient joints lie in a common hinge plane orthogonal with respect to the spindle longitudinal axis.

3. The linear drive according to claim 1, wherein the resilient joints are each configured as solid state hinges positioned between two sliding sleeve segments which are adjacent to one another in the spindle peripheral direction and each solid state hinge consists of a bar section of the sliding sleeve which is connected in one piece to the two adjacent sliding sleeve segments and configured to be twisted resiliently.

4. The linear drive according to claim 1, wherein the sliding sleeve is longitudinally slotted at positions spaced apart from one another in the spindle peripheral direction to form the sliding sleeve segments.

5. The linear drive according to claim 1, wherein the sliding sleeve is made of a plastic.

6. The linear drive according to claim 1, wherein the sliding sleeve segments are configured in a rocker-like manner and configured to be swiveled in a rocker-like manner about the associated swivel area in a direction radial to the spindle longitudinal axis due to the resilient joints, wherein each sliding sleeve segment comprises two segment wings extending from its swivel area in opposite axial directions, at least one of which segment wings forms a support wing provided with one of the support area sections.

7. The linear drive according to claim 6, wherein of the two segment wings of each sliding sleeve segment, only one segment wing forms a support wing, while the other segment wing is configured as a fastening wing which is spaced apart from the inner peripheral surface of the hollow output rod and contributes to fixing the sliding sleeve to the rolling bearing.

8. The linear drive according to claim 7, wherein for each sliding sleeve segments which are adjacent in the spindle peripheral direction, the axial orientation of the two segment wings is interchanged in such a way that their support wings point in axially opposite directions starting from the swivel area and the sliding sleeve comprises two segment wing circles positioned axially next to each other with support wings and fastening wings positioned alternately in succession in the spindle peripheral direction.

9. The linear drive according to claim 1, wherein the sliding sleeve comprises on its radial inner circumference an annular seating area which is coaxial with the spindle longitudinal axis and points radially inwards and with which annular seating area the sliding sleeve is seated on a radial outer peripheral surface of an outer ring of the rolling bearing fixed with an inner ring to the threaded spindle.

10. The linear drive according to claim 9, wherein the resilient joints of the sliding sleeve are at the same axial height as the annular seating area, wherein the sliding sleeve segments project axially beyond the annular seating area.
  11. The linear drive according to claim 9, wherein the annular seating area of the sliding sleeve comprises a plurality of segment seating area sections which are positioned in succession in the spindle peripheral direction and are each formed on the radially inner surface of one of the sliding sleeve segments and which segment seating area sections are each axially bounded on one side by a retaining surface, which is formed on a retaining projection formed on the associated fastening wing and projecting radially inwards and with which retaining surface the associated sliding sleeve segment bears axially against one of the two axial end faces of the outer ring of the rolling bearing, so that the outer ring is held between retaining surfaces positioned at a distance from one another in the spindle peripheral direction and facing axially towards the segment seating area sections.
  12. The linear drive according to claim 11, wherein the annular seating area of the sliding sleeve comprises a plurality of hinge seating area sections each positioned between two segment seating area sections positioned in succession in the spindle peripheral direction, which hinge seating area sections are formed on the radial inner surface of the resilient joints.
  13. The linear drive according to claim 1, wherein each sliding sleeve segment has a segment longitudinal axis which lies in a swivel plane of the sliding sleeve segment overlapping with the spindle longitudinal axis and which segment longitudinal axis is inclined with respect to the spindle longitudinal axis, the segment longitudinal axes having an opposite inclination in the spindle peripheral direction of adjacent sliding sleeve segments.
  14. The linear drive according to claim 13, wherein the segment longitudinal axis of each sliding sleeve segment has a greater inclination before insertion of the sliding sleeve into the rod cavity than in the position of use of the sliding sleeve inserted into the rod cavity and supported on the inner peripheral surface of the output rod.
  15. The linear drive according to claim 1, wherein each sliding sleeve segment has a radial segment outer surface facing the inner peripheral surface of the output rod, wherein the support area sections of the sliding sleeve segments are each formed by a section of the radial segment outer surface of the associated sliding sleeve segment and wherein each segment outer surface is convexly curved at least in the region of the support area section both in the spindle peripheral direction and in the axial direction of the spindle longitudinal axis.
  16. The linear drive according to claim 1, wherein the sliding sleeve segments are each positioned on the rolling bearing so as to be tiltable with respect to the rolling bearing in order to enable their radial swiveling.
  17. The linear drive according to claim 1, wherein a rotary drive apparatus is positioned at the rear of the drive housing.
  18. The linear drive according to claim 17, wherein the rotary drive apparatus is an electric rotary drive apparatus.
  19. The linear drive according to claim 15, wherein each segment outer surface is convexly curved in its entirety both in the spindle peripheral direction and in the axial direction of the spindle longitudinal axis.
  20. The linear drive according to claim 18, wherein the rotary drive apparatus is formed by an electric motor.
-